MDMIS

Mineral Detection & Mining Intelligence System (MDMIS)

Software Project Management Report

Version 1.0
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1. Executive Summary

The **Mineral Detection & Mining Intelligence System (MDMIS)** is an integrated mining intelligence platform designed to support the complete mining lifecycle — from mineral exploration through to export compliance. The system combines Artificial Intelligence (AI), Geographic Information Systems (GIS), satellite imagery, unmanned aerial vehicles (drones), Ground Penetrating Radar (GPR), compliance management, and predictive analytics into a single unified platform.

This Software Project Management Plan (SPMP) defines the project objectives, scope, human and technical resources, schedules, risk register, mitigation strategies, quality standards, communication protocols, and project monitoring and control approaches that will govern the successful delivery of the MDMIS platform.

2. Project Objectives

2.1 Primary Objectives

- Detect mineral deposits using AI-driven sensor fusion techniques.
- Improve the accuracy and efficiency of mining resource estimation.
- Monitor mining operations in real-time through IoT and telemetry data.
- Predict and proactively mitigate safety risks on mine sites.
- Ensure full regulatory compliance throughout the mining lifecycle.
- Provide end-to-end mineral traceability from extraction to export.
- Support transportation logistics and inventory management operations.
- Generate intelligent reports, dashboards, and executive decision-support tools.

2.2 Business Objectives

- Reduce exploration costs by a minimum of 30% through AI-assisted targeting.
- Increase mineral recovery rates by at least 20% via optimised extraction planning.
- Significantly reduce the incidence of illegal and artisanal mining activities.
- Improve the speed and accuracy of compliance reporting to regulatory bodies.
- Increase overall profitability of mining operations through data-driven decision making.

3. Project Scope

3.1 In Scope

Module	Key Components
Exploration Module	Satellite data analysis, drone data integration, GPR, electromagnetic sensors
AI Module	Mineral classification, concentration estimation, prediction models, risk assessment
GIS Module	Geological mapping, mineral distribution maps, mine planning support
Operations Module	Inventory management, transportation tracking, production monitoring
Compliance Module	Government & OECD reporting, traceability management
Analytics Module	Executive dashboards, financial analytics, safety analytics

3.2 Out of Scope

- Physical drone manufacturing or hardware procurement.
- Sensor hardware production or certification.
- Automation of physical mining equipment.
- Design or implementation of autonomous drilling systems.

4. Project Organization Structure

The following roles and responsibilities govern the MDMIS project team:

Role	Key Responsibility
Project Sponsor	Strategic oversight, funding approval, executive escalation
Project Manager	Day-to-day project control, schedule, budget, risk management
Business Analyst (x2)	Requirements elicitation, stakeholder liaison, use-case documentation
System Architect	Solution architecture, technology stack decisions, integration design
GIS Specialist (x2)	Geological mapping, spatial data management, GIS platform setup
AI Engineer (x3)	Machine learning model development, training, and evaluation
Backend Developer (x4)	REST API development, microservices, database integration
Frontend Developer (x3)	User interface design and implementation, dashboard development
Database Administrator	Database schema design, performance tuning, data governance
QA Engineer (x3)	Test planning, execution, defect tracking, UAT coordination
DevOps Engineer	CI/CD pipelines, cloud infrastructure, containerisation
Security Analyst	Security architecture, penetration testing, compliance auditing
End Users	User acceptance testing, feedback, operational readiness

5. Software Development Methodology

5.1 Agile Scrum Framework

MDMIS will be developed using the **Agile Scrum** methodology with two-week sprint cycles. This approach enables iterative delivery, continuous stakeholder feedback, early identification of risks, and the flexibility to adapt to evolving requirements throughout the project lifecycle.

Sprint Cycle Activities

1. Sprint Planning — define goals, allocate user stories, estimate effort.
2. Requirement Analysis — clarify acceptance criteria with stakeholders.
3. Design — produce technical designs for sprint deliverables.
4. Development — implement features and integrations.
5. Testing — unit, integration, and regression testing.
6. Sprint Review — demonstrate completed work to stakeholders.
7. Retrospective — identify improvements for the next sprint.

Key Benefits

- Faster time-to-value through incremental feature delivery.
- Continuous stakeholder engagement and course correction.
- Improved risk management via short feedback cycles.
- Transparent progress tracking using velocity and burndown metrics.

6. Work Breakdown Structure (WBS)

Phase 1 — Initiation

- Identify and onboard key stakeholders.
- Conduct feasibility study and technology assessment.
- Complete initial business requirements analysis.

Phase 2 — Planning

- Define detailed project scope and acceptance criteria.
- Develop resource allocation and procurement plan.
- Perform comprehensive risk assessment and register.

Phase 3 — Design

- Produce system and solution architecture documentation.
- Complete database schema and data dictionary design.
- Deliver UI/UX wireframes and prototypes for stakeholder sign-off.

Phase 4 — Development

- Module 1: Exploration Intelligence — satellite, drone, GPR integration.
- Module 2: GIS Mapping — geological maps, spatial analysis tools.
- Module 3: AI Classification Engine — mineral detection and prediction models.
- Module 4: Compliance Management — government and OECD reporting workflows.
- Module 5: Traceability System — chain-of-custody tracking from extraction to export.
- Module 6: Reporting & Analytics — dashboards, financial and safety analytics.

Phase 5 — Testing

- Unit Testing — individual component verification.
- Integration Testing — end-to-end workflow validation.
- Security Testing — penetration testing and vulnerability assessment.
- Performance Testing — load, stress, and scalability testing.
- User Acceptance Testing (UAT) — sign-off by end users.

Phase 6 — Deployment

- Deploy to cloud infrastructure (AWS/Azure) with multi-region configuration.
- Conduct structured user training and onboarding sessions.
- Execute go-live and hypercare support plan.

Phase 7 — Maintenance

- Ongoing bug triage and resolution.
- Feature enhancement based on user feedback and regulatory updates.
- Continuous system monitoring and performance optimisation.

7. Project Schedule

Phase	Duration	Cumulative Week
Initiation	2 Weeks	Wk 1–2
Planning	2 Weeks	Wk 3–4
Design	4 Weeks	Wk 5–8
Development	16 Weeks	Wk 9–24
Testing	6 Weeks	Wk 25–30
Deployment	2 Weeks	Wk 31–32
Training	2 Weeks	Wk 33–34
Maintenance	Ongoing	Wk 35+

Total Planned Duration: 34 Weeks (approximately 8.5 calendar months), excluding ongoing maintenance activities.

8. Resource Management Plan

8.1 Human Resources

Role	Headcount	Engagement
Project Manager	1	Full-time, project duration
Business Analysts	2	Full-time, Phases 1–4
GIS Experts	2	Full-time, Phases 3–5
AI Engineers	3	Full-time, Phases 3–6
Backend Developers	4	Full-time, Phases 3–6
Frontend Developers	3	Full-time, Phases 3–6
QA Engineers	3	Full-time, Phases 4–6
DevOps Engineer	1	Full-time, project duration
Security Analyst	1	Part-time, Phases 2–6

8.2 Technical Infrastructure & Tools

Category	Technology / Tool
Cloud Platform	AWS / Microsoft Azure (multi-region)
Database	PostgreSQL with PostGIS spatial extension
AI / ML Frameworks	Python, TensorFlow, PyTorch, scikit-learn
Backend Services	Node.js, RESTful APIs, GraphQL
Frontend Framework	React.js, Tailwind CSS
Containerisation	Docker, Kubernetes (K8s)
GIS Platform	GeoServer, QGIS, Esri ArcGIS
DevOps & CI/CD	GitHub Actions, Jenkins, Terraform
Monitoring	Prometheus, Grafana, ELK Stack

9. Risk Management Plan

The following risk assessment matrix identifies the key risks to the MDMIS project, their likelihood of occurrence, their potential business impact, and the resulting priority level assigned for mitigation planning.

Risk	Probability	Impact	Priority
Sensor Data Failure	High	High	**CRITICAL**
AI Prediction Inaccuracy	Medium	High	**HIGH**
Security Breach	Medium	High	**HIGH**
Scope Creep	High	Medium	**HIGH**
Budget Overrun	Medium	High	**HIGH**
Regulatory Changes	Medium	Medium	**MEDIUM**
Cloud Service Failure	Low	High	**MEDIUM**
Staff Turnover / Shortage	Medium	Medium	**MEDIUM**

10. Mitigation Plan & Strategies

Risk 1: Sensor Data Failure

Impact: Potential for incorrect or incomplete mineral detection, leading to flawed exploration decisions.

Mitigation Strategies

- Deploy multi-sensor redundancy so no single point of failure exists.
- Implement real-time sensor health monitoring with automated alerting.
- Schedule automated calibration routines and maintenance windows.
- Maintain backup sensor arrays and alternative data feeds.

Contingency Plan: Revert to curated satellite imagery archives and historical geological datasets pending sensor restoration.

Risk 2: AI Model Inaccuracy

Impact: Erroneous mineral classification or concentration estimates may misdirect investment decisions.

Mitigation Strategies

- Curate large, high-quality, geologically diverse training datasets.
- Enforce continuous model retraining pipelines with version control.
- Apply k-fold cross-validation and ensemble model techniques.
- Require expert geologist sign-off before AI outputs influence operational decisions.

Contingency Plan: Engage independent geological consultants for manual review and validation of critical outputs.

Risk 3: Security Breach

Impact: Unauthorised access could result in data theft, reputational damage, and compliance violations.

Mitigation Strategies

- Enforce multi-factor authentication (MFA) for all user accounts.
- Implement role-based access control (RBAC) with least-privilege principles.
- Apply end-to-end encryption for data in transit and at rest.
- Conduct quarterly third-party security audits and penetration tests.

Contingency Plan: Activate incident response team immediately; isolate affected systems; initiate disaster recovery protocol.

Risk 4: Scope Creep

Impact: Uncontrolled scope expansion may cause schedule delays and cost overruns.

Mitigation Strategies

- Establish a formal Change Control Board (CCB) to evaluate all change requests.
- Freeze requirements at the start of each sprint to prevent mid-sprint disruption.
- Require written stakeholder approval for any scope additions.

Contingency Plan: Re-prioritise the product backlog; defer non-critical features to future release phases.

Risk 5: Budget Overrun

Impact: Unanticipated expenditure may threaten project funding continuity.

Mitigation Strategies

- Conduct monthly budget reviews against earned value metrics.
- Use Agile incremental delivery to enable early spend visibility.
- Optimise resource utilisation through cross-functional staffing.

Contingency Plan: Invoke phased implementation strategy; negotiate deferred scope with project sponsor.

Risk 6: Regulatory Changes

Impact: New or amended mining regulations may require significant compliance module rework.

Mitigation Strategies

- Monitor mining legislation and OECD guidelines on a continuous basis.
- Design compliance module with modular, configuration-driven architecture.

Contingency Plan: Issue rapid-turnaround compliance update releases via dedicated regulatory sprint.

Risk 7: Cloud Infrastructure Failure

Impact: Cloud outages may cause system-wide downtime and data inaccessibility.

Mitigation Strategies

- Deploy across multiple geographic cloud regions for high availability.
- Schedule automated backups with tested restore procedures.
- Implement load balancing and auto-scaling to absorb traffic spikes.

Contingency Plan: Activate secondary disaster recovery site; failover within defined RTO/RPO targets.

Risk 8: Staff Turnover

Impact: Departure of key personnel may result in knowledge gaps and delivery delays.

Mitigation Strategies

- Enforce comprehensive documentation standards for all deliverables.
- Conduct regular knowledge transfer sessions and pair programming.
- Maintain cross-trained team members for all critical roles.

Contingency Plan: Activate pre-identified backup resources or specialist contractors without delay.

11. Quality Management Plan

11.1 Quality Objectives

Quality Metric	Target Standard
System Availability (Uptime)	≥ 99.9% (excluding scheduled maintenance)
AI Model Prediction Accuracy	≥ 95% validated against test datasets
API & UI Response Time	< 3 seconds under normal load conditions
Critical Security Vulnerabilities	Zero at go-live; patched within 24 hours post-discovery
Defect Escape Rate (to Production)	< 2% of total defects identified
User Acceptance Rate	≥ 90% approval from end-user stakeholders

11.2 Quality Assurance Activities

- Mandatory peer code reviews for all pull requests prior to merge.
- Automated unit and regression test suites integrated into the CI/CD pipeline.
- Quarterly independent security audits and penetration testing exercises.
- Load and stress performance testing prior to each major release.
- Structured User Acceptance Testing (UAT) with defined pass/fail criteria.

12. Communication Management Plan

Stakeholder Group	Communication Method	Frequency	Owner
Project Sponsor	Executive Progress Report	Monthly	Project Manager
Project Team	Scrum Stand-up Meeting	Daily	Scrum Master
Senior Management	Status Dashboard Report	Bi-weekly	Project Manager
Clients / Users	Sprint Review & Demo	End of Sprint	Product Owner
Regulators	Formal Compliance Report	Quarterly	Compliance Lead
Risk Committee	Risk Register Review	Monthly	Project Manager

13. Change Management Plan

All changes to the project scope, schedule, budget, or technical architecture must follow the formal Change Request process outlined below. No change may be implemented without written approval from the Change Control Board (CCB).

Change Request Workflow

Step 1: **Submit Change Request** — Requestor documents the proposed change, rationale, and expected benefits.

Step 2: **Impact Analysis** — Project Manager and Architect assess schedule, technical, and resource impacts.

Step 3: **Cost Analysis** — Finance lead estimates cost implications and budget variance.

Step 4: **CCB Approval / Rejection** — Change Control Board reviews findings and issues a formal decision.

Step 5: **Implementation** — Approved changes are scheduled into the appropriate sprint backlog.

Step 6: **Testing & Verification** — QA team validates that the change meets acceptance criteria.

Step 7: **Release & Communication** — Change is deployed and all affected stakeholders are notified.

14. Monitoring & Control

14.1 Key Performance Indicators (KPIs)

KPI	Description	Target
Schedule Performance Index (SPI)	Earned value vs planned schedule progress	≥ 0.95
Cost Performance Index (CPI)	Earned value vs actual cost expenditure	≥ 0.95
Defect Density	Defects per KLOC per sprint	< 2 / KLOC
System Uptime	Platform availability percentage	≥ 99.9%
AI Prediction Accuracy	Model accuracy on validation datasets	≥ 95%
User Satisfaction Score	Post-UAT satisfaction survey result	≥ 4.0 / 5.0
Compliance Report Automation	Percentage of reports auto-generated	≥ 95%

14.2 Reporting Cadence

- Weekly sprint progress reports distributed to the core project team.
- Bi-weekly status dashboard shared with senior management.
- Monthly executive progress briefing presented to the Project Sponsor.
- Quarterly stakeholder review including risk register and KPI assessment.

15. Project Success Criteria

The MDMIS project will be formally declared successful upon achievement of all of the following criteria, validated through independent review and stakeholder sign-off:

- Total project cost delivered within ±10% of the approved baseline budget.
- All planned deliverables completed within the agreed 34-week project schedule.
- AI mineral classification accuracy validated at ≥ 95% on independent test data.
- Platform uptime performance maintained at ≥ 99.9% during the go-live stabilisation period.
- User acceptance survey rating of ≥ 90% approval from operational end users.
- Demonstrated improvement in mining profitability of at least 20% within 12 months post-launch.
- Compliance reporting automation rate of ≥ 95% for all applicable regulatory submissions.
- Zero unresolved critical security vulnerabilities identified at the point of go-live.

16. Conclusion

The Mineral Detection & Mining Intelligence System (MDMIS) represents a technically sophisticated, economically compelling, and highly scalable advancement in mining technology. By integrating Artificial Intelligence, Geographic Information Systems, drone and satellite remote sensing, Ground Penetrating Radar, blockchain-ready traceability, regulatory compliance automation, and executive-level analytics into a single unified ecosystem, MDMIS addresses the full spectrum of challenges faced by modern mining operations.

This Software Project Management Plan provides the structured governance framework necessary to deliver the platform successfully. Through disciplined Agile Scrum methodology, proactive risk management, rigorous quality assurance, and continuous stakeholder engagement, the project team is well-positioned to deliver MDMIS on time, within budget, and to the quality standards expected by all stakeholders.

Upon successful deployment, MDMIS is projected to materially reduce mineral exploration costs, increase resource recovery rates, strengthen regulatory compliance, enhance mine-site safety, and deliver measurable improvements in operational profitability — fully realising the strategic vision articulated in the original MDMIS business case and feasibility documentation.